

NELSON MANDELA

UNIVERSITY

STERILE PRODUCTS MANUFACTURE

Pharmaceutical Science III

ZTM301

N Worthington

Outline

- General principles
- Controlling microbial contamination
- Facility design
- Atmosphere
- Personnel
- Attire
- Equipment

Outline

- Starting materials
- Manufacturing processes:
 - General principles
 - Terminally sterilised products
 - Aseptic preparation
- Case Studies

General Principles

- Quality of a product cannot be ensured solely by examining in detail a small number of units
 - Low levels of microorganisms may not be detected
- Products must be manufactured in a suitable environment by a procedure that minimises the possibility of contamination.
- Tests can be performed at the end as an additional safeguard.
- Sterile products manufacture is subject to special requirements.
 - minimise risks of contamination

General Principles

- Much depends on the skill, training and attitudes of the personnel involved.
- Manufacture must strictly follow carefully established and validated methods.

Controlling Microbial Contamination

- Manufacture should be carried out in clean areas
- Entry should be through airlocks (personnel, equipment and materials)
- Clean areas should be maintained to an appropriate cleanliness standard
- Supplied with air which has passed through HEPA filters.
- Component preparation, product preparation and filling should be carried out in separate areas within the clean area.

Facility Design

- M.O may be transferred to a product from working surfaces, fixtures and equipment.
- Pooled stagnant water is a frequent source of contamination. (especially gram -ve MO's)
- All premises, should be in a clean, dry and tidy condition.
- All exposed surfaces in clean rooms should be smooth, impervious and unbroken
 - minimise the shedding/accumulation of particles or micro-organisms
 - and to permit repeated application of cleaning agents and disinfectants

Facility Design

- Coving should be used at the junction between walls and floors or ceilings.
- There should be no uncleanable recesses.
- Minimally projecting shelves & equipment.
- Pipes, ducts and utilities should not create recesses, unsealed openings and surfaces which are difficult to clean.
- Sinks and drains should be prohibited in grade A/B areas.
- Floor drains in lower grade clean rooms should be fitted with traps or water seals to prevent back-flow.

- Why are sinks and drains a problem?

Facility Design

- Changing rooms should be designed as airlocks
 - Provide physical separation of the different stages of changing
- Minimise microbial and particulate contamination of protective clothing
- Hand washing facilities should be provided only in the first stage of the changing rooms.
- Both airlock doors should not be opened simultaneously.
- Filtered air supply should maintain a positive pressure
- Increase in Grade is stepwise

Atmosphere

- Clean air is a prerequisite during manufacture.
 - M.O. may be suspended in the atmosphere
- Microbial count of air may be reduced by filtration, chemical disinfection and UV light.
- Chemical disinfectants are limited in their use
 - irritant properties when sprayed.
- UV irradiation at wavelengths between 280 and 240nm
 - effective at short distances from the source
 - moulds are often resistant to this treatment
- Filtration is the most commonly used method.

Atmosphere

Grade A	Grade B	Grade C and D
• <u>Zone for high risk operations</u> • Filling zone, open ampoules and vials • Conditions provided by a laminar air flow work station. • Provide a homogeneous air speed in a range of 0,36 – 0,54 m/s at the working position. • The maintenance of laminarity should be demonstrated and validated.	• <u>Background environment for grade A zone</u> • For aseptic preparation and filling	• Clean areas for carrying out less critical stages in the manufacture of sterile products • preparation of solutions.

Atmosphere (Non-viable)

	At rest		In operation	
Grade	Maximum permitted number of particles/m³ equal to or above			
	0,5 µm	5 µm	0.5 µm	5 µm
A	3 500	1	3 500	1
B	3 500	1	350 000	2 000
C	350 000	2 000	3 500 000	20 000
D	3 500 000	20 000	not defined	not defined

Atmosphere (Viable)

Recommended limits for microbial contamination

Grade	Air sample, cfu/m ³	Settle plates (diam. 90 mm), cfu/4 hours	Contact plates (diam. 55 mm), cfu/plate	Glove print, 5 fingers, cfu/glove
A	< 1	< 1	< 1	< 1
B	10	5	5	5
C	100	50	25	-
D	200	100	50	-

Clean Room Classification

Class	0.1 μm	0.2 μm	0.3 μm	0.5 μm	1 μm	5 μm
1	35	7	3	1		
10	350	75	30	10	1	
100		750	300	100	10	1
1000				1000	100	10
10000				10000	1000	100
100000				100000	10000	10000
<i>Measure in particles/ft³</i>						

Personnel

- Personnel can transfer potentially harmful organisms to a product
- High standards of hygiene are therefore very important
- Operators should be free from communicable disease
- Should have no open lesions on the exposed body surfaces.
- Hand washing facilities and protective garments must be provided.
- Direct contact between the materials and the operator's hands must be avoided.

Personnel

- Personnel manufacturing sterile products should receive basic training in microbiology.
- All personnel should receive regular training in disciplines relevant to the correct manufacture of sterile products
- Periodic health checks for such conditions should be conducted
- Changing and washing should follow a written procedure.
- Wristwatches, make-up and jewellery should not be worn in clean areas.
- The clothing and its quality should be appropriate for the process and the grade of the working area.

Clean Room Attire: Grade A/B

- Headgear should totally enclose hair (beard and moustache)
- It should be tucked into the neck of the suit
- A face mask should be worn to prevent the shedding of droplets.
- Appropriate sterilised, non-powdered rubber or plastic gloves and sterilised or disinfected footwear should be worn.
- Trouser-bottoms should be tucked inside the footwear and garment sleeves into the gloves.
- The protective clothing should shed virtually no fibres and retain particles shed by the body.

Clean Room Attire: Grade C/D

Grade C

- Hair (beard and moustache) should be covered.
- A single or two-piece trouser suit, gathered at the wrists and with high neck and appropriate shoes or overshoes should be worn.
- They should shed virtually no fibres or particulate matter.

Grade D

- Hair and beard should be covered.
- A general protective suit and shoes or overshoes should be worn.

Clean Room Attire

- Outdoor clothing should not be brought into changing rooms leading to grade B and C rooms.
- Every worker in a grade A/B area should be provided with clean sterile protective garments at each work session.
- Gloves should be regularly disinfected during operations

Equipment

- All equipment should be easy to dismantle and clean.
- It should be inspected for cleanliness before use.
- Maintenance and repairs should be carried out outside the clean area.
- If sterilisation is required it should be carried out after complete reassembly.
- When equipment maintenance has been carried out within the clean area, the area should be cleaned, disinfected and sterilised before processing recommences.
 - if the required standards of cleanliness have not been maintained during the work.

Filling Equipment: Glass Ampoules



Ampoule Filling

www.ReAgent.co.uk

Blow/fill/seal technology

- Purpose built machines
- In one continuous operation, containers are formed from a thermoplastic granulate, filled and then sealed, all by the one automatic machine.
- Fitted with an effective grade A air shower
- May be installed in at least a grade C environment, provided that grade A/B clothing is used.
- The environment should comply with the viable and non-viable limits “at rest”
- Viable limit only when in operation.

Equipment

- equipment used for the production of terminally sterilised products should be installed in at least a grade D environment.
- Particular attention should be paid to:
 - equipment design and qualification,
 - validation and reproducibility of cleaning-in-place and sterilisation-in-place,
 - background clean room environment in which the equipment is located,
 - operator training and clothing,
 - interventions in the critical zone of the equipment including any aseptic assembly prior to the commencement of filling.

Starting Materials: Quality

- Raw materials are an important source of m.o.
- Materials of natural origin contain levels of m.o.
- Materials with a low bioburden assists in the control of
 - the quality of the final product and
 - contamination in the environment.
- The possibility of product spoilage by non-pathogenic microorganisms must be assessed.
 - growing in the presence of a preservative

Starting Materials: Quality

- Effectiveness of an in-process treatment to remove or destroy them must be evaluated.
- Where possible starting materials of natural origin may be sterilized:
 - either before use or
 - during the manufacturing process.
- Heat treatment, filtration, irradiation and recrystallisation from a bactericidal solvent such as alcohol may also be used.

Starting Materials: Water

- The microbial content of water is extremely important
 - It forms a major constituent of many products
 - Is used for washing equipment and for cooling processes.
- The main concerns are:
 - the quality of the raw water
 - the purifying processing it receives
 - the holding and distribution systems.
 - biofilm

Starting Materials: Water

Raw or mains water:

- The quality of the raw water varies with the source of supply, and the treatment given.
- M.O. in fresh water are nutritionally undemanding
- Have a relatively low optimum growth temperature.
- Surface water has more abundant and faster growing flora than water from deep sources.
- 90% of m.o. of stored industrial water supplies were found to be Gram negative bacteria. (Endotoxins!)
- Other organisms include yeast and fungi.

Starting Materials: Water

- Unless water in storage tanks is treated, the count may rise to 10^6 microorganisms per ml.
- Softened water is often used
 - for washing containers or
 - in cooling systems required for semisolid preparations.
- Soft water is prepared by
 - a base exchange method
 - by a lime soda ash process or
 - by addition of sodium hexametaphosphate.
- The chemical filter beds may act as a reservoir for microflora.

Water for Injections

- As it leaves the still, distilled water is free from microorganisms.
- Subsequent contamination occurs:
 - because of a fault in the cooling system
 - in the storage vessel or the distribution system
- It is used after distillation with a specially designed still,.
- A post-distillation heat sterilisation stage is commonly included in the process.
- It is often stored, recirculating at 80 °C.

Deionised Water

- De-ionized water is prepared by passing mains water through anion and cation exchange resin beds
- Remove ions.
- Bacteria present in the mains water are still present in the de-ionised water.
- Beds which are not regenerated frequently are often heavily contaminated
- This has prompted the development of resins able to resist microbiological contamination.

Deionised Water

- A large-pore, strong base, macroreticular, quarternary ammonium anion exchange resin.
- Permits m.o. to enter the pore cavity
- Electrostatically binds them to the cavity surface.
- Deionised water is used in pharmaceutical formulations for washing containers and plant & for the preparation of disinfectant solutions and in laboratories

Reverse Osmosis Water (RO)

- Water is forced by an osmotic pressure through a semipermeable membrane.
- Acts as a molecular filter.
- The diffusion of solutes dissolved in the water is impeded
- The process removes microorganisms and their pyrogens.
- Molecules most difficult to remove are small inorganic ones like sodium chloride

Reverse Osmosis Water (RO)

- Passage through two membranes in series is used
 - to increase the efficiency of removal of these molecules
 - to decrease the risk of structural failure of a membrane
- Membranes are usually composed of cellulose ester or polyamides.

Water Storage & Distribution

- The rate of production of WFI is usually not sufficient to meet processing demands
- Collected in a holding tank for subsequent use.
- The storage vessel may act as a microbial reservoir
- NB that contents of all storage vessels are tested regularly.
- In large operations the holding tanks may be part of a continuously operating system.

Water Storage & Distribution

- Water should be re-sterilized just prior to return to the main storage tank.
 - membrane filtration or UV light treatment
- In such instances it should be held at a temperature too high for microbial growth
 - usually 80 °C.
- WFI may be stored at room temperature but for a maximum of 24 hr.
- Collected as a batch for a particular use with any unused water being discarded within 24hr.

Water Storage & Distribution

- System requires frequent sanitisation.
- The stainless-steel storage tanks are connected to a welded stainless steel distribution loop.
- The tank is provided with a hydrophobic membrane vent filter.
 - capable of excluding bacteria and nonviable particulate matter.
- When the water cannot be used at 80 °C, heat exchangers must be installed to reduce the temperature at the point of use.

Biofilm

- A biofilm comprises any group of microorganisms in which cells stick to each other and often also to a surface
- These adherent cells become embedded within a slimy extracellular matrix that is composed of extracellular polymeric substances (EPS)
- The cells within the biofilm produce the EPS components, which are typically a polymeric conglomeration of extracellular polysaccharides, proteins, lipids and DNA
- Because they have three-dimensional structure and represent a community lifestyle for microorganisms, they have been metaphorically described as "cities for microbes".

Biofilm

- Common in water storage systems if required conditions are not adhered
- Difficult to eradicate because the film protects the microorganisms
- May slough-off in clumps
- Prevent by having smooth inner surface of piping, no dead-legs

Sterile Manufacturing: General Principles

- Precautions to minimise contamination should be taken during all processing stages.
- Reduction of **bioburden** in paramount
- Activities in clean areas should be kept to a minimum.
- Movement of personnel should be controlled and methodical.
- avoid excessive shedding of particles and organisms

General Principles

- Products of microbial origin should not be made in areas with other products except:
 - Vaccines consisting of dead microorganisms, microbial extracts or inactivated viruses
 - The completeness of inactivation must be proven before processing.
- Separate premises are needed for:
 - filling of live or attenuated vaccines
 - production of biological medicinal products derived from live organisms.

Holding Times

- Holding times should be minimised and subject to a time limit appropriate to the storage conditions:
 - Washing, drying, sterilisation and use of components/containers and equipment
- Time between the preparation of a solution and its sterilisation or filtration should be minimal.
- Holding times should take into account a products composition and the prescribed method of storage.
- Holding times must be validated

Preparation Methods

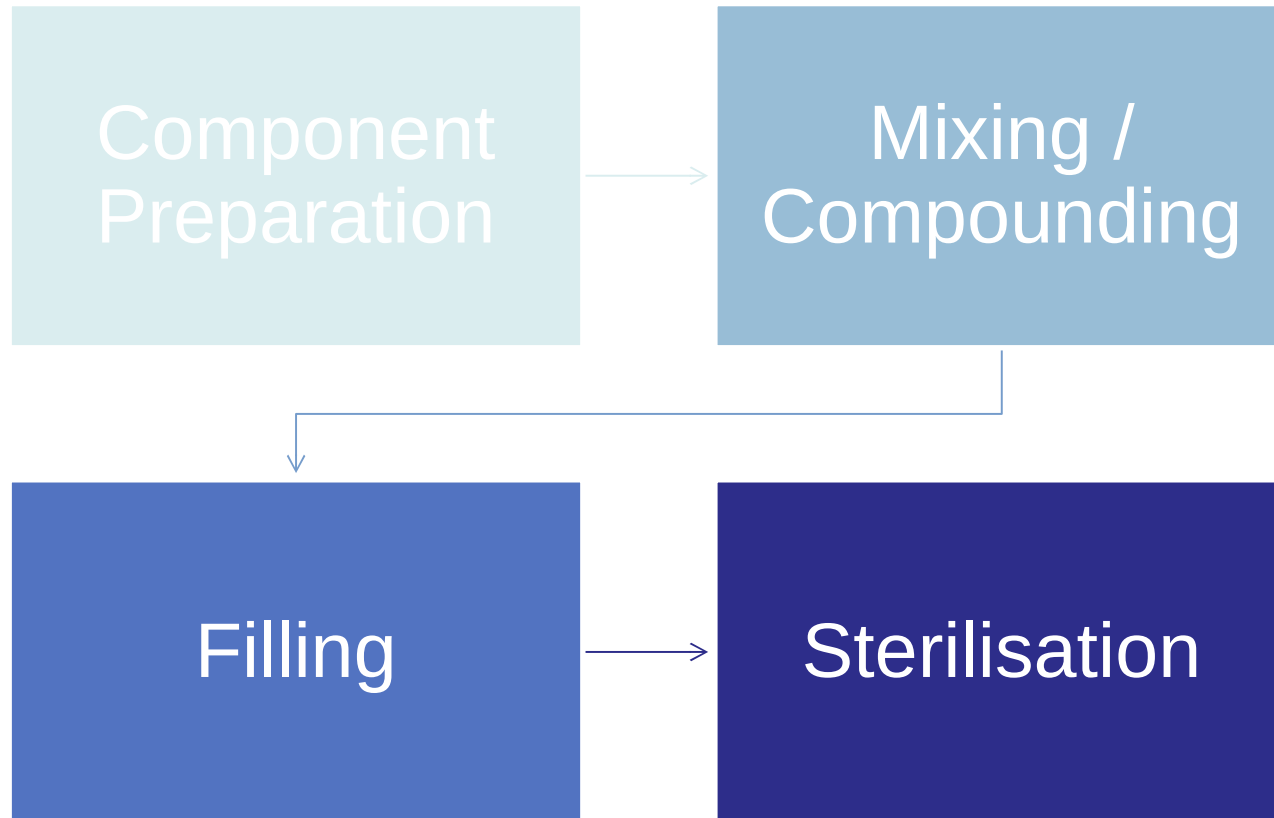
Products may be prepared according to two primary processes:

- Terminal sterilisation
 - Used for products that withstand the harsh conditions of sterilisation
- Aseptic transfer
 - Required for products that are sensitive to temperature or other forms of sterilisation and can't be filter sterilised.

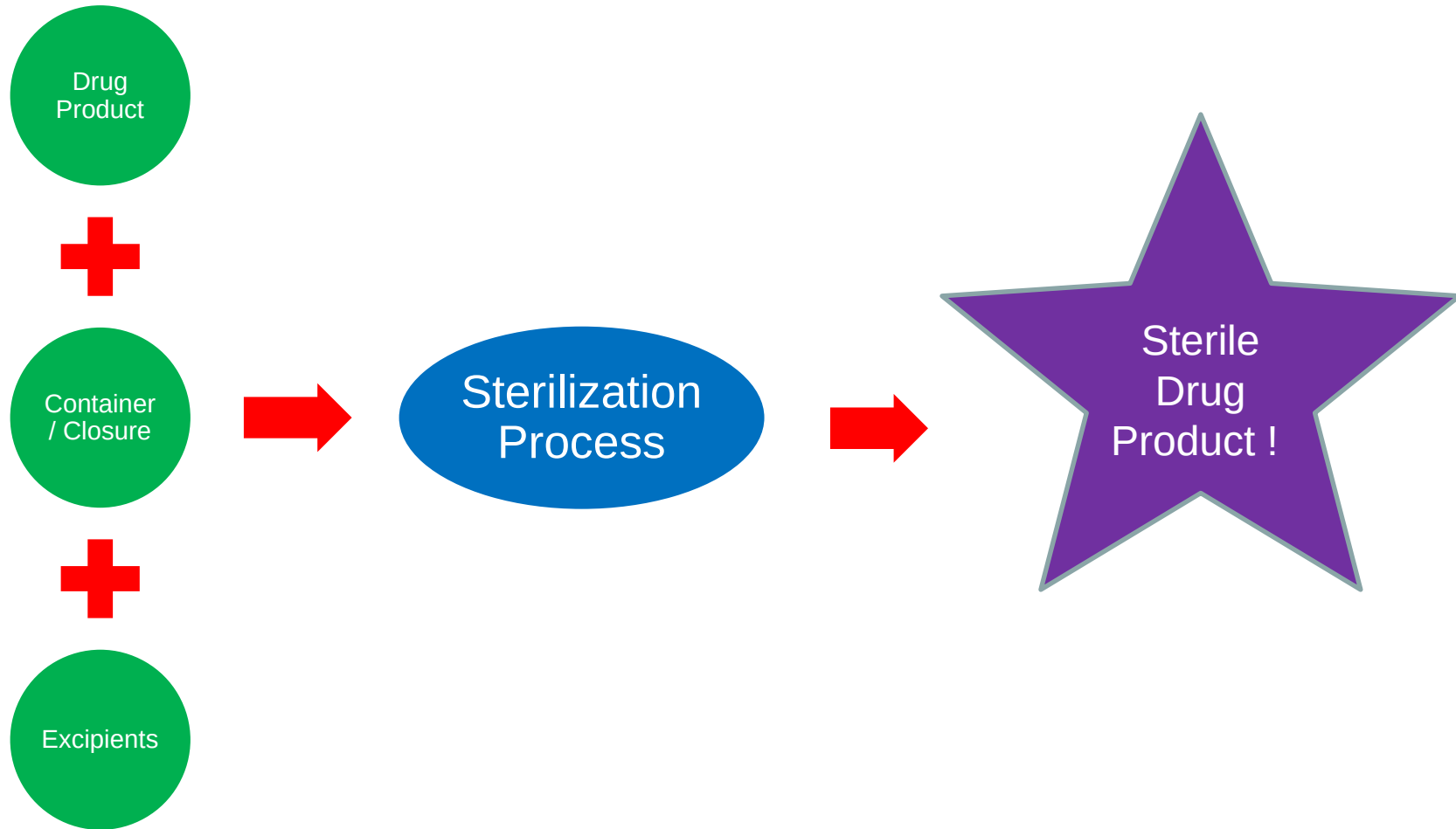
Decision Tree for Aseptic Products

- <http://academy.gmp-compliance.org/guidemgr/files/WC500204724.PDF>
- Also on Moodle
- Familiarise yourself with decision trees

Terminal Sterilisation



Terminal Sterilization



Sterilization Process must be compatible with all components !

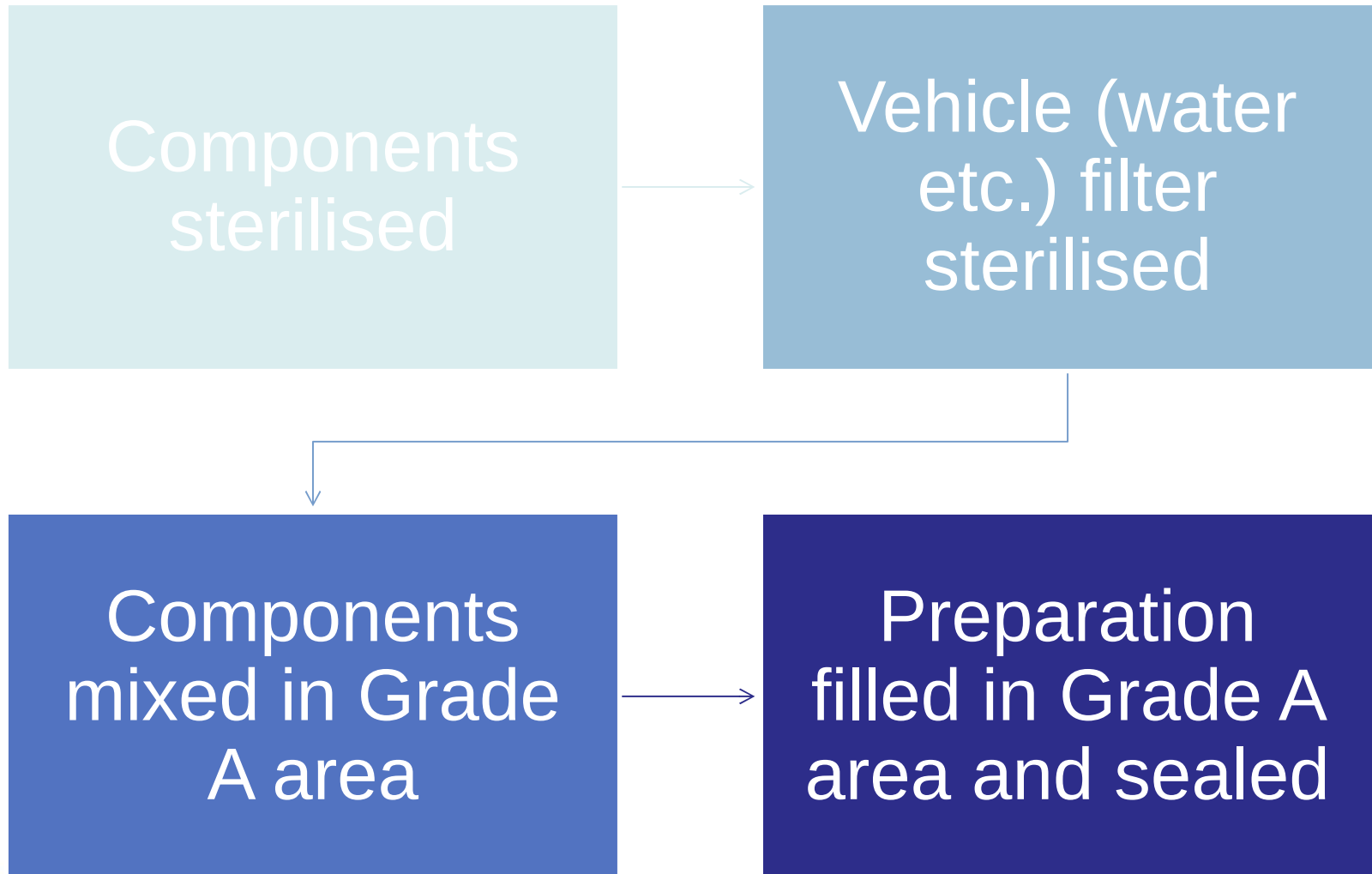
Terminal Sterilisation

- After preparation, the product is transferred to containers, sealed and then immediately sterilised (Dry/Moist heat, radiation or ethylene oxide).
- Preparation of components and most products should be done in at least a grade D environment.
- Where there is unusual microbial risk (product actively supports m.o growth) preparation should be done in a grade C environment.

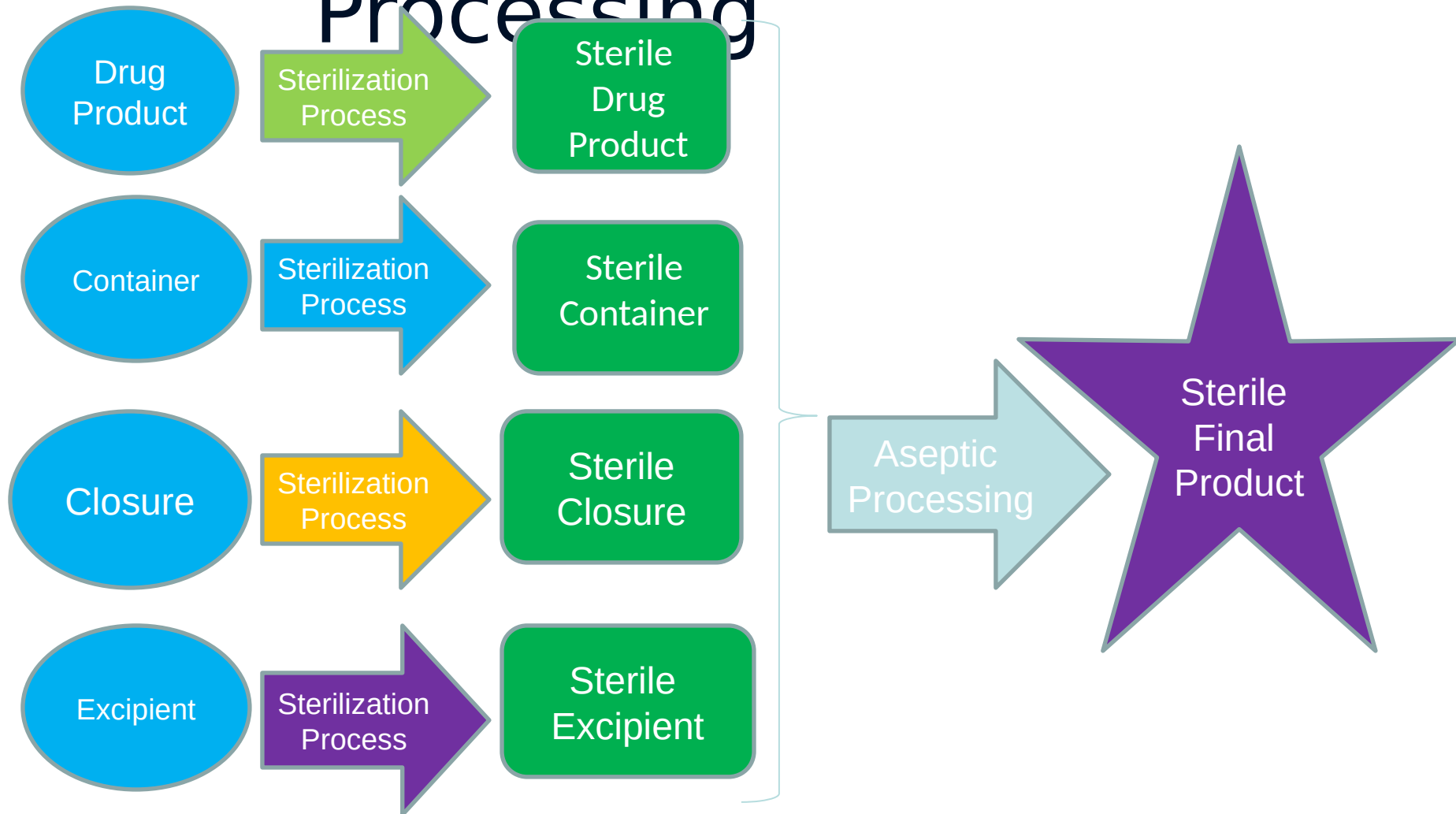
Terminal Sterilisation

- Filling done in grade C zone.
- Where the product is at unusual risk of contamination from the environment (i.e. filling operation is slow or containers are wide-necked)
 - Filling done in grade A with at least grade C background
- Preparation and filling of ointments, creams, suspensions and emulsions should be done in a grade C environment before terminal sterilisation.

Aseptic Transfer



Aseptic Processing



Can use multiple sterilization processes each optimized for the individual component

Aseptic Transfer

- Prepared from previously sterilised materials or by filtration sterilisation and filled into sterile containers
- Components after washing should be handled in at least a grade D environment.
- Handling of sterile starting materials and components, should be done in a grade A environment with grade B background.

Aseptic Transfer

- Solutions that are sterile filtered during the process should be prepared in a grade C environment.
- Mixtures that can't be filtered should be prepared in a grade A environment with a grade B background.
- Handling & filling : grade A with grade B background.

Aseptic Transfer

- Preparation and filling of sterile ointments/creams, suspensions and emulsions should be done in a grade A environment, with a grade B background.
- if the product is exposed and is not subsequently filtered

The Four Pillars of a Robust Aseptic Process

- Personnel training & monitoring
- Environmental monitoring
- Facilities design & HVAC validation
- Process simulation (media fills)

Case Studies

Design a process to manufacture the following products:

1. 1 ml atropine ampoules
2. 10 ml soluble insulin multi-dose vial
3. 5 ml medroxyprogesterone acetate multidose vial

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